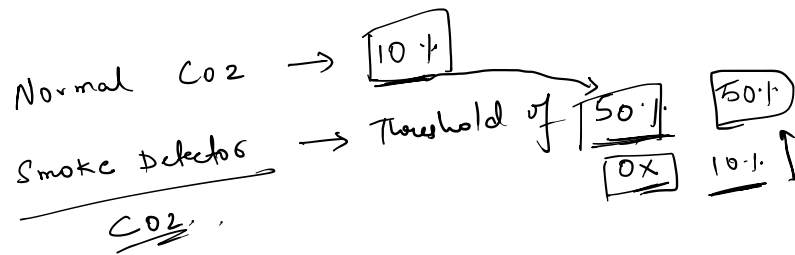
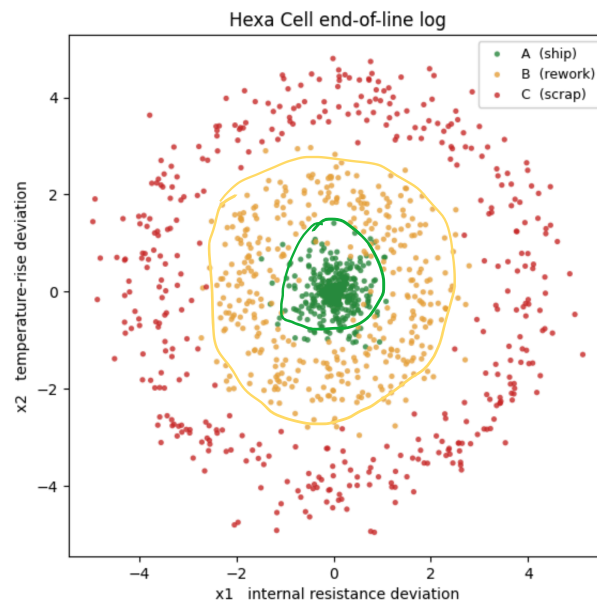


Ground Rules:

- 1) The class will start at 9:05 PM.
- 2) Break around 10:00 PM.
- 3) Class will end by 11:15 + don't leaving
- 4) Generic questions in question tab.



* Hexa Cell: Data Distribution.



- 1) Multi Class
- 2) Non linearly separable

1) Logistic Regression Model:

- 1) Can it handle multi class??

- By default No.

linear data?

2) Can it handle non-linear?
- By default No.

2) KNN :

1) Can it handle multi class?

- By default yes

2) Can it handle Non linear data?

- By default Yes

Problem : It is hard to handle high dimensional with KNN.

$x_3 \rightarrow$ Image of the cell

1024

$m \times n$

$\rightarrow 1024 \times 1024$
 $\approx 10^6$

3) Decision Trees

1) Multi class?

Yes

2) Non linear data?

Yes.

Problem : Decision Trees do not work well with sparse data.

4) SVM :

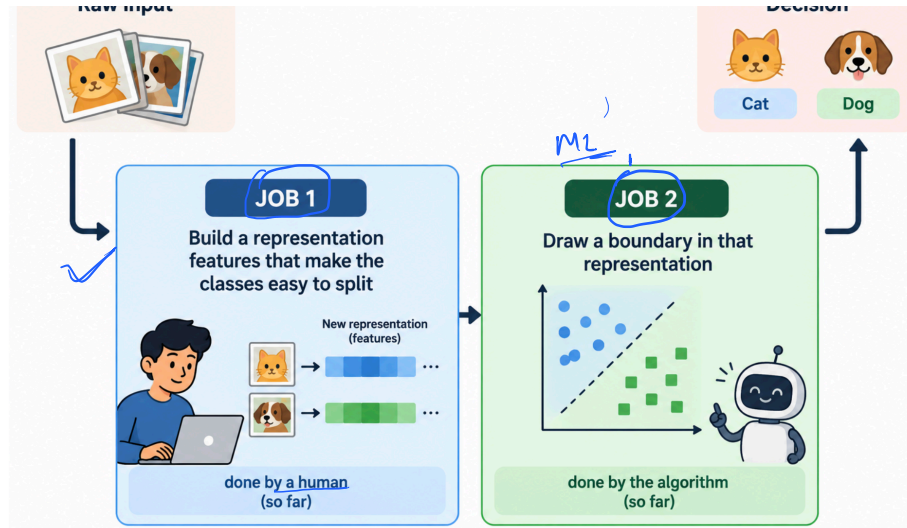
1) Multi class?

By default No

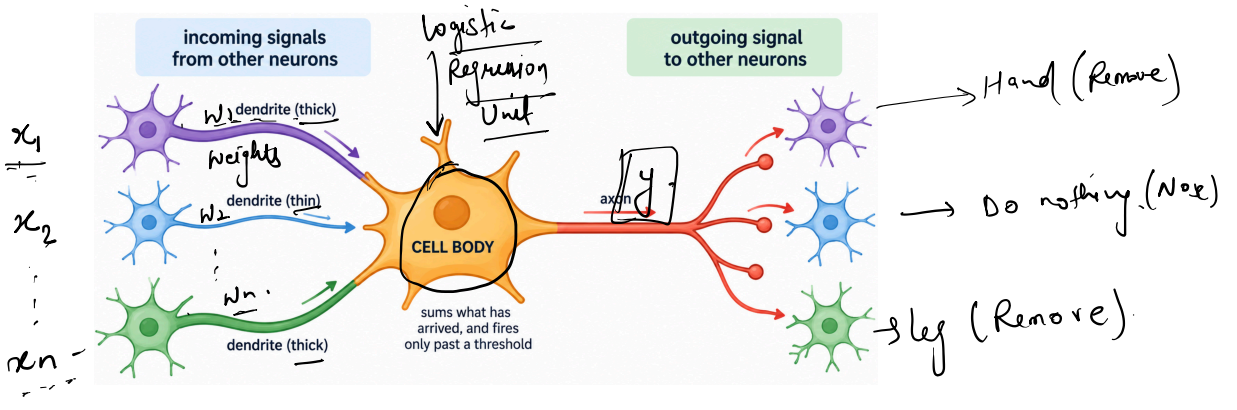
2) Non linear data :

Yes, but with kernel trick.

Input →

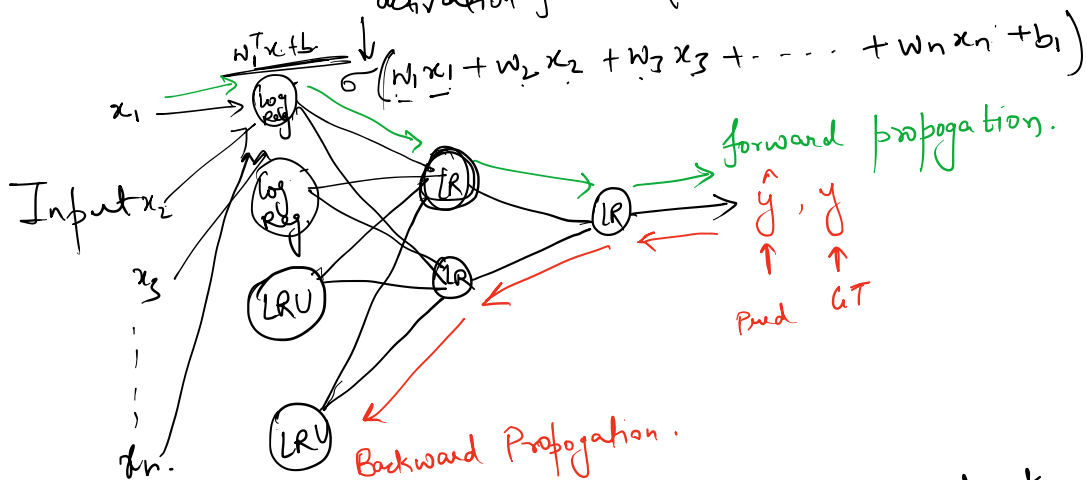


Iron Rod
(Hand)
(Nose)
(leg)



ETA Prediction → weather, Traffic, Driver availability, Distance, Type of food, time of the day, Queue at Resto, Open Order.

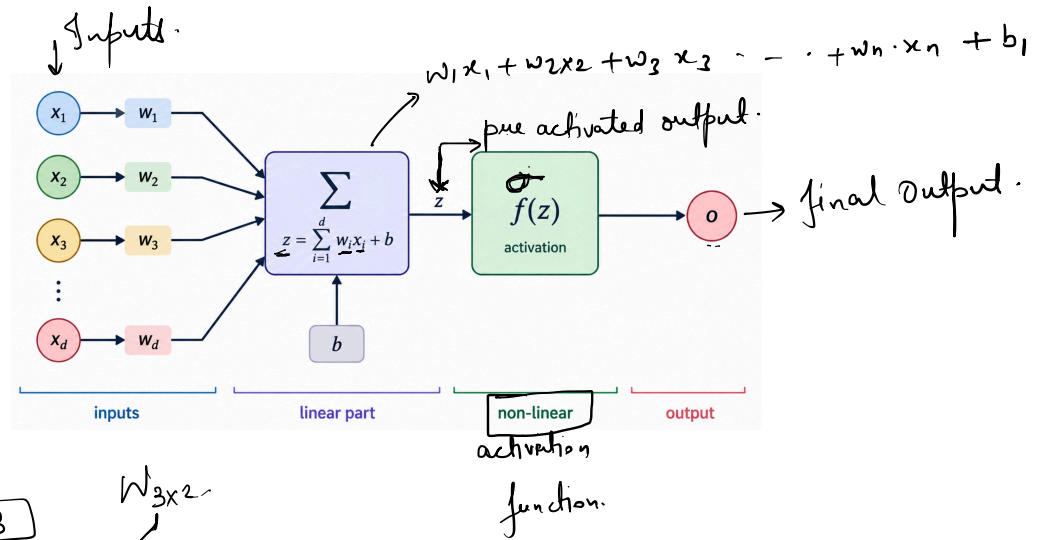
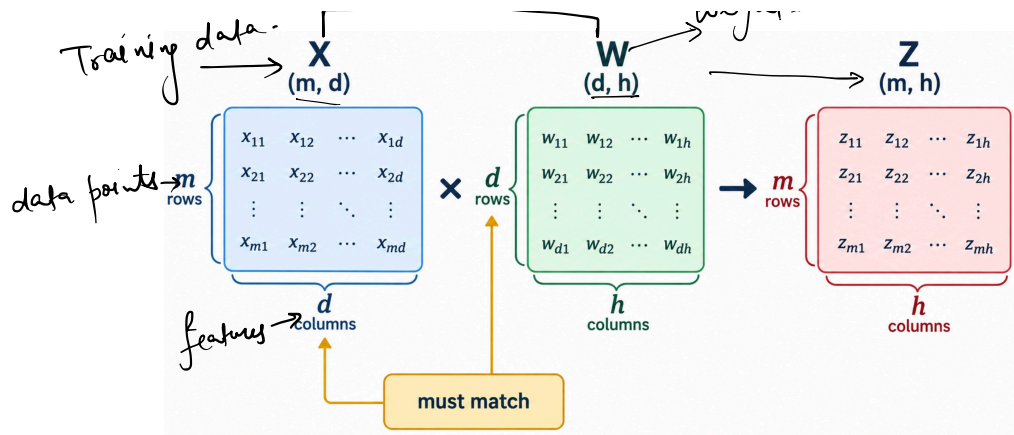
activation function. (σ , tanh, ReLU, ...)



Network of Neurons → Neural Network.

(1200, 3)
m d

neighbor



1Mx2

